Oracle Spatial Project.

Spatial database - firefighter base

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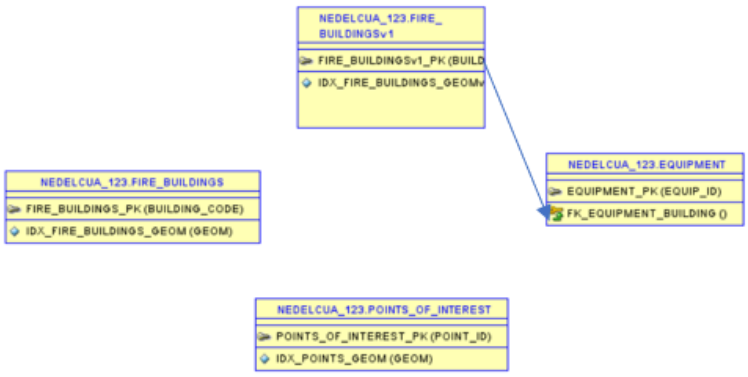
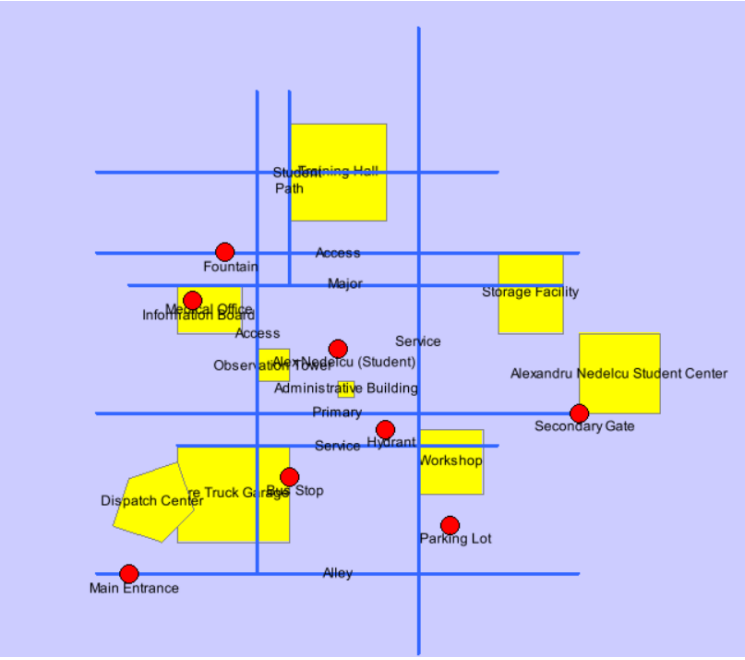
1. Conceptual schema of the database



2. Drawing with spatial objects



3. Scripts for creating spatial tables

```
// fire buildings (polygons)
CREATE TABLE FIRE_BUILDINGS (
  building_code VARCHAR2(25) PRIMARY KEY,
  name VARCHAR2(100),
  address VARCHAR2(100),
  geom SDO_GEOMETRY
);
// roads (lines)
CREATE TABLE ROADS (
  road_id NUMBER PRIMARY KEY,
  name VARCHAR2(100),
  category VARCHAR2(50),
  geom SDO_GEOMETRY
);
// points of interest (points)
CREATE TABLE POINTS_OF_INTEREST (
  point_id NUMBER PRIMARY KEY,
  name VARCHAR2(100),
  category VARCHAR2(50),
  geom SDO_GEOMETRY
);
// equipment (non-spatial)
CREATE TABLE EQUIPMENT (
  equip_id NUMBER PRIMARY KEY,
  equip_name VARCHAR2(100),
  building_code VARCHAR2(25),
  status VARCHAR2(30)
);
```

I forgot to add a connection between Buildings and Equipment so I will add it now.

```
SELECT building_code
FROM EQUIPMENT
WHERE building_code NOT IN (SELECT building_code FROM FIRE_BUILDINGS);
ALTER TABLE EQUIPMENT
ADD CONSTRAINT fk_equipment_building
FOREIGN KEY (building_code)
REFERENCES FIRE_BUILDINGS(building_code);
SELECT constraint_name, table_name, status
FROM user_constraints
WHERE table_name = 'EQUIPMENT';
```

4. Scripts for adding spatial data

Buildings

```
INSERT INTO FIRE_BUILDINGS VALUES (
 'B1','Fire Truck Garage','Fire Rd. 10',
 MDSYS.SDO_GEOMETRY(2003,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,1003,3),
  MDSYS.SDO_ORDINATE_ARRAY(5,2, 12,8))
);
INSERT INTO FIRE_BUILDINGS VALUES (
 'B2','Dispatch Center','Fire Rd. 11',
 MDSYS.SDO_GEOMETRY(2003,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,1003,1),
  MDSYS.SDO_ORDINATE_ARRAY(1,3, 4,2, 6,4, 5,7, 2,6, 1,3))
);
INSERT INTO FIRE_BUILDINGS VALUES (
```

```
 'B3','Observation Tower','Fire Rd. 12',
 MDSYS.SDO_GEOMETRY(2003,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,1003,3),
  MDSYS.SDO_ORDINATE_ARRAY(10,12, 12,14))
);
INSERT INTO FIRE_BUILDINGS VALUES (
 'B4','Administrative Building','Fire Rd. 13',
 MDSYS.SDO_GEOMETRY(2003,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,1003,3),
  MDSYS.SDO_ORDINATE_ARRAY(15,11, 16,12))
);
INSERT INTO FIRE_BUILDINGS VALUES (
 'B5','Workshop','Workshop Rd. 1',
 MDSYS.SDO_GEOMETRY(2003,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,1003,3),
  MDSYS.SDO_ORDINATE_ARRAY(20,5, 24,9))
);
INSERT INTO FIRE_BUILDINGS VALUES (
 'B6','Storage Facility','Storage Rd. 2',
 MDSYS.SDO_GEOMETRY(2003,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,1003,3),
  MDSYS.SDO_ORDINATE_ARRAY(25,15, 29,20))
);
INSERT INTO FIRE_BUILDINGS VALUES (
 'B7','Medical Office','Health Rd. 5',
 MDSYS.SDO_GEOMETRY(2003,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,1003,3),
  MDSYS.SDO_ORDINATE_ARRAY(5,15, 9,18))
);
INSERT INTO FIRE_BUILDINGS VALUES (
 'B8','Training Hall','Training Rd. 3',
```

```
 MDSYS.SDO_GEOMETRY(2003,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,1003,1),
  MDSYS.SDO_ORDINATE_ARRAY(12,22, 18,22, 18,28, 12,28, 12,22))
);
INSERT INTO FIRE_BUILDINGS VALUES (
 'B9','Alexandru Nedelcu Student Center','Student Rd. 1',
 MDSYS.SDO_GEOMETRY(2003,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,1003,3),
  MDSYS.SDO_ORDINATE_ARRAY(30,10, 35,15))
);
```

Roads

```
INSERT INTO ROADS VALUES (
 1,'Main Alley','Alley',
 MDSYS.SDO_GEOMETRY(2002,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,2,1),
  MDSYS.SDO_ORDINATE_ARRAY(0,0, 30,0))
);
INSERT INTO ROADS VALUES (
 2,'Side Road','Service',
 MDSYS.SDO_GEOMETRY(2002,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,2,1),
  MDSYS.SDO_ORDINATE_ARRAY(20,-5, 20,34))
);
INSERT INTO ROADS VALUES (
 3,'Boulevard','Primary',
 MDSYS.SDO_GEOMETRY(2002,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,2,1),
  MDSYS.SDO_ORDINATE_ARRAY(0,10, 30,10))
);
```

```
INSERT INTO ROADS VALUES (
 4,'Access Street','Access',
 MDSYS.SDO_GEOMETRY(2002,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,2,1),
  MDSYS.SDO_ORDINATE_ARRAY(0,20, 30,20))
);
INSERT INTO ROADS VALUES (
 5,'North-South Route','Access',
 MDSYS.SDO_GEOMETRY(2002,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,2,1),
  MDSYS.SDO_ORDINATE_ARRAY(10,0, 10,30))
);
INSERT INTO ROADS VALUES (
 6,'Connector','Service',
 MDSYS.SDO_GEOMETRY(2002,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,2,1),
  MDSYS.SDO_ORDINATE_ARRAY(5,8, 25,8))
);
INSERT INTO ROADS VALUES (
 7,'Bypass','Major',
 MDSYS.SDO_GEOMETRY(2002,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,2,1),
  MDSYS.SDO_ORDINATE_ARRAY(2,18, 29,18))
);
INSERT INTO ROADS VALUES (
 8,'Training Track','Path',
 MDSYS.SDO_GEOMETRY(2002,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,2,1),
  MDSYS.SDO_ORDINATE_ARRAY(12,30, 12,18))
);
INSERT INTO ROADS VALUES (
```

```
 9,'Alexandru Nedelcu Avenue','Student',
 MDSYS.SDO_GEOMETRY(2002,NULL,NULL,
  MDSYS.SDO_ELEM_INFO_ARRAY(1,2,1),
  MDSYS.SDO_ORDINATE_ARRAY(0,25, 25,25))
);
```

Points of interest

```
INSERT INTO POINTS_OF_INTEREST VALUES (
 1,'Main Entrance','Gate',
 MDSYS.SDO_GEOMETRY(2001,NULL,MDSYS.SDO_POINT_TYPE(2,0,NULL),NULL,NULL)
);
INSERT INTO POINTS_OF_INTEREST VALUES (
 2,'Fountain','Decoration',
 MDSYS.SDO_GEOMETRY(2001,NULL,MDSYS.SDO_POINT_TYPE(8,20,NULL),NULL,NULL)
);
INSERT INTO POINTS_OF_INTEREST VALUES (
 3,'Parking Lot','Facility',
 MDSYS.SDO_GEOMETRY(2001,NULL,MDSYS.SDO_POINT_TYPE(22,3,NULL),NULL,NULL)
);
INSERT INTO POINTS_OF_INTEREST VALUES (
 4,'Secondary Gate','Gate',
 MDSYS.SDO_GEOMETRY(2001,NULL,MDSYS.SDO_POINT_TYPE(30,10,NULL),NULL,NULL)
);
INSERT INTO POINTS_OF_INTEREST VALUES (
 5,'Bus Stop','Transport',
 MDSYS.SDO_GEOMETRY(2001,NULL,MDSYS.SDO_POINT_TYPE(12,6,NULL),NULL,NULL)
);
INSERT INTO POINTS_OF_INTEREST VALUES (
 6,'Information Board','Info',
 MDSYS.SDO_GEOMETRY(2001,NULL,MDSYS.SDO_POINT_TYPE(6,17,NULL),NULL,NULL)
```

```
);
INSERT INTO POINTS_OF_INTEREST VALUES (
 7,'Hydrant','Infrastructure',
 MDSYS.SDO_GEOMETRY(2001,NULL,MDSYS.SDO_POINT_TYPE(18,9,NULL),NULL,NULL)
);
INSERT INTO POINTS_OF_INTEREST VALUES (
 8,'Alex Nedelcu (Student)','Student',
 MDSYS.SDO_GEOMETRY(2001,NULL,MDSYS.SDO_POINT_TYPE(15,14,NULL),NULL,NULL)
);
```

Equipment

```
INSERT INTO EQUIPMENT VALUES (1,'Engine 1','B1','Operational');
INSERT INTO EQUIPMENT VALUES (2,'Ladder 2','B1','Operational');
INSERT INTO EQUIPMENT VALUES (3,'Generator A','B5','Maintenance');
INSERT INTO EQUIPMENT VALUES (4,'Water Pump','B6','Operational');
INSERT INTO EQUIPMENT VALUES (5,'Medical Kit','B7','Available');
INSERT INTO EQUIPMENT VALUES (6,'Training Dummy','B8','Available');
INSERT INTO EQUIPMENT VALUES (7,'Hydrant Valve Kit','B3','Available');
INSERT INTO EQUIPMENT VALUES (8,'Alex Nedelcu - Personal Kit','B4','Personal');
```

5. Updating the spatial metadata table

```
INSERT INTO USER_SDO_GEOM_METADATA (
 TABLE_NAME, COLUMN_NAME, DIMINFO, SRID
) VALUES (
 'FIRE_BUILDINGS','GEOM',
 MDSYS.SDO_DIM_ARRAY(
   MDSYS.SDO_DIM_ELEMENT('X',0,100,0.05),
   MDSYS.SDO_DIM_ELEMENT('Y',0,100,0.05)
```

```
 ), NULL
);
INSERT INTO USER_SDO_GEOM_METADATA (
 TABLE_NAME, COLUMN_NAME, DIMINFO, SRID
) VALUES (
 'ROADS','GEOM',
 MDSYS.SDO_DIM_ARRAY(
   MDSYS.SDO_DIM_ELEMENT('X',0,100,0.05),
   MDSYS.SDO_DIM_ELEMENT('Y',0,100,0.05)
 ), NULL
);
INSERT INTO USER_SDO_GEOM_METADATA (
 TABLE_NAME, COLUMN_NAME, DIMINFO, SRID
) VALUES (
 'POINTS_OF_INTEREST','GEOM',
 MDSYS.SDO_DIM_ARRAY(
   MDSYS.SDO_DIM_ELEMENT('X',0,100,0.05),
   MDSYS.SDO_DIM_ELEMENT('Y',0,100,0.05)
 ), NULL
);
```

6. Creating spatial indexes

```
CREATE INDEX idx_fire_buildings_geom ON FIRE_BUILDINGS(geom) INDEXTYPE IS
MDSYS.SPATIAL_INDEX;

CREATE INDEX idx_roads_geom ON ROADS(geom) INDEXTYPE IS MDSYS.SPATIAL_INDEX;

CREATE INDEX idx_points_geom ON POINTS_OF_INTEREST(geom) INDEXTYPE IS
MDSYS.SPATIAL_INDEX;
```

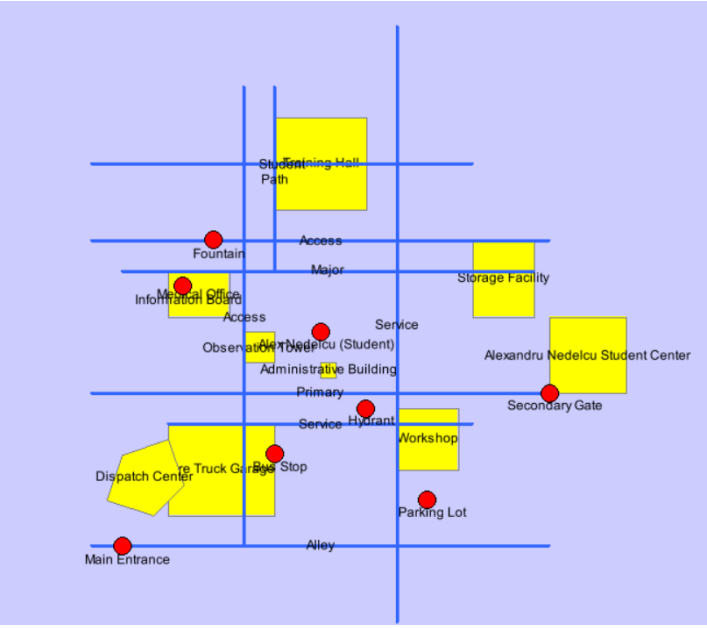
7. Visualizing spatial data in Map View, under various criteria

SELECT BUILDING_CODE, NAME, GEOM FROM FIRE_BUILDINGS

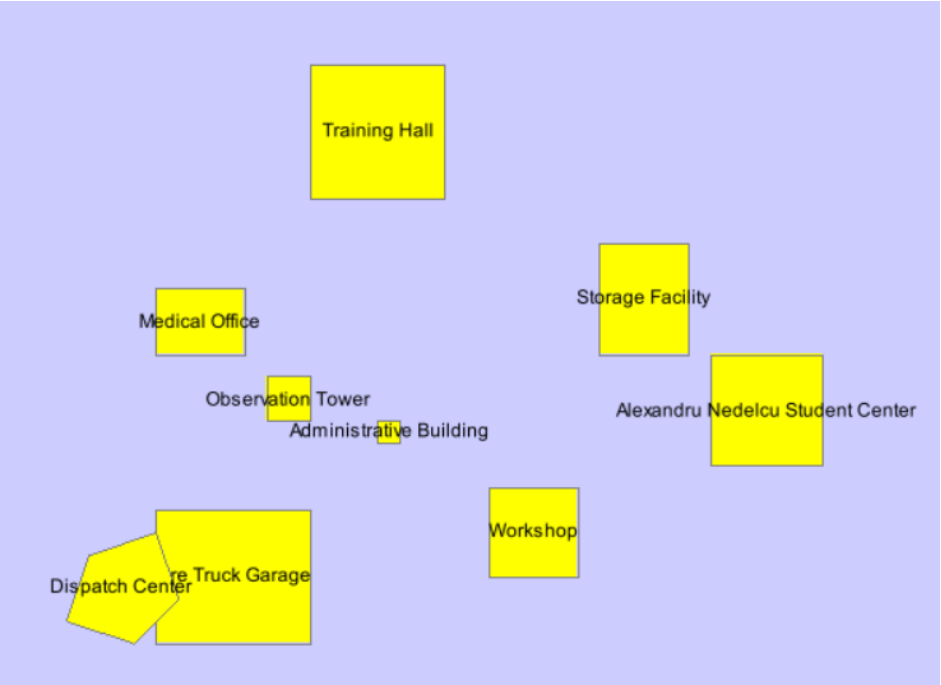
SELECT ROAD_ID, CATEGORY, GEOM FROM ROADS

SELECT POINT_ID, NAME, CATEGORY, GEOM FROM POINTS_OF_INTEREST

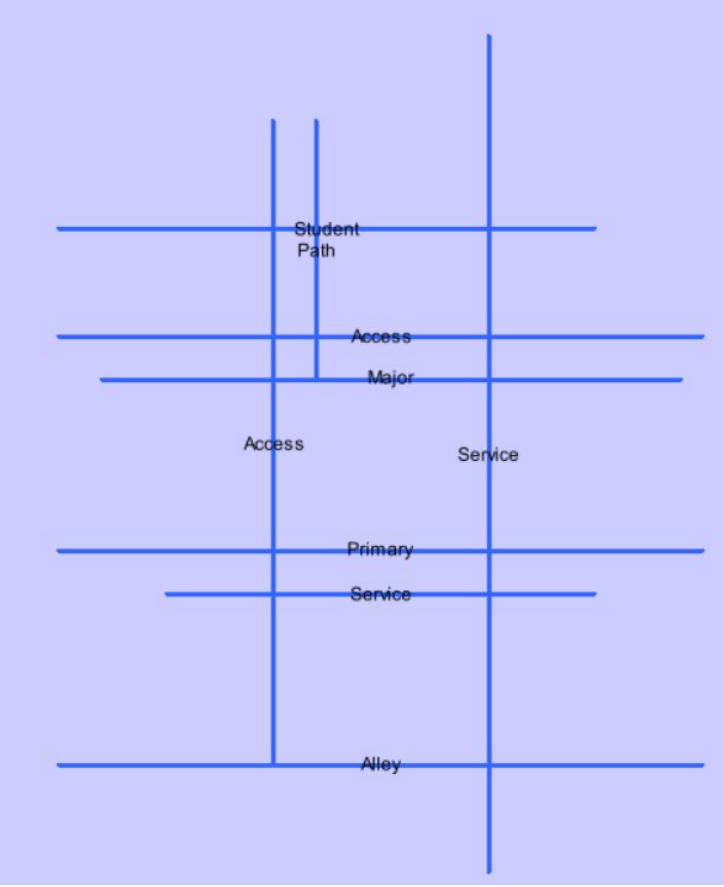
All 3 spatial layers



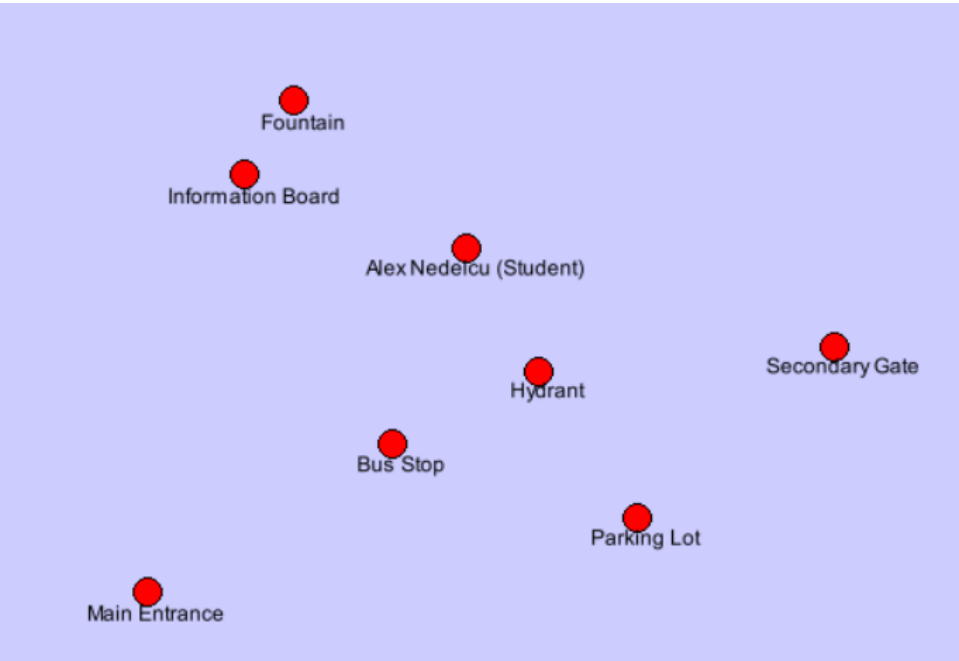
Buildings layer only



Roads layer only



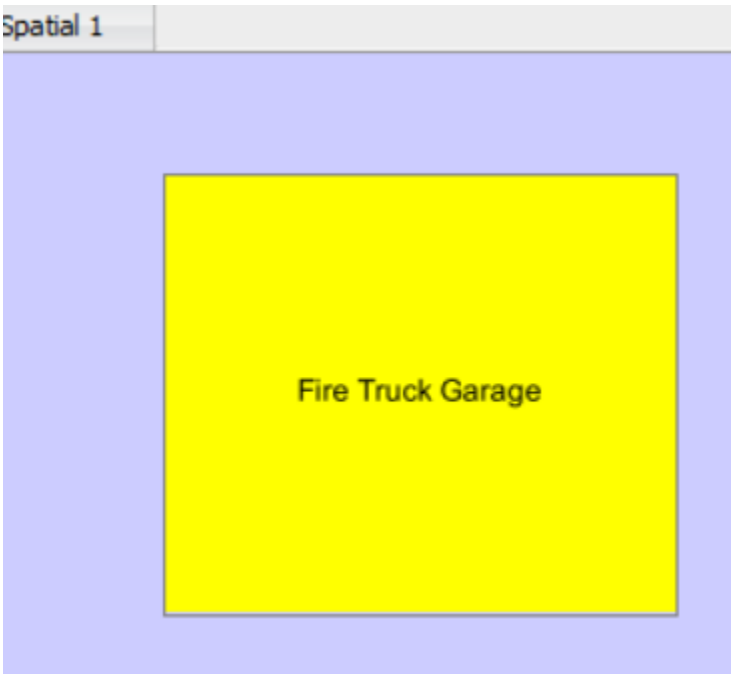
Point of interets layer only



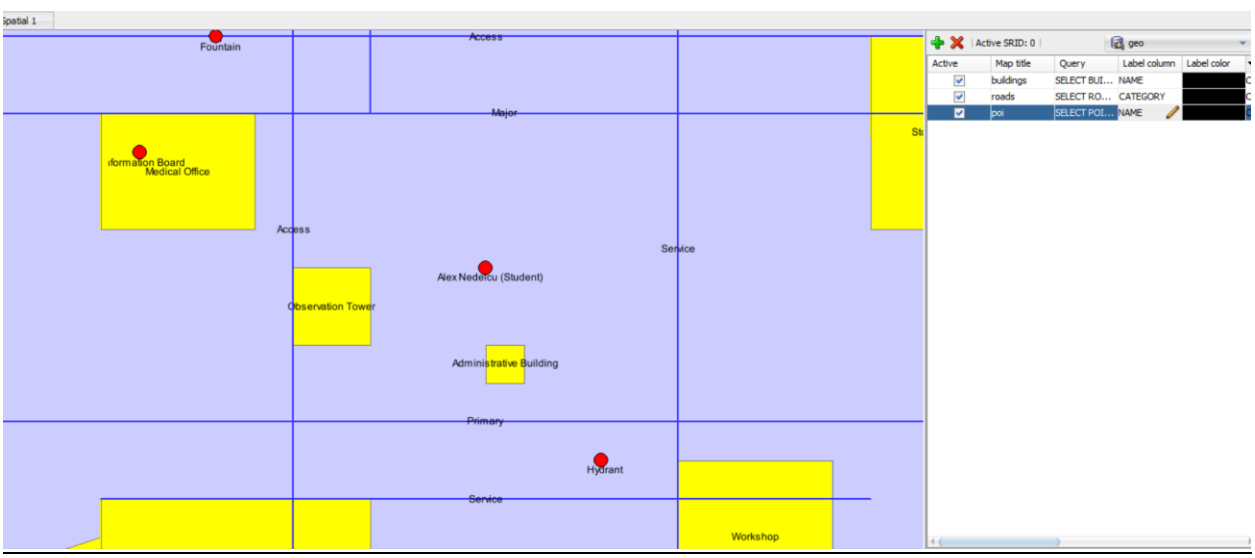
For some reasons, i cannot use the filter option in Map View so I used directly the code.

Only the garage

SELECT BUILDING_CODE, NAME, GEOM FROM FIRE_BUILDINGS WHERE NAME = 'Fire Truck Garage'



Zoomed in



8. Spatial queries (requirements, SQL and PL-SQL code) that should include other spatial functions than those exemplified in the seminar

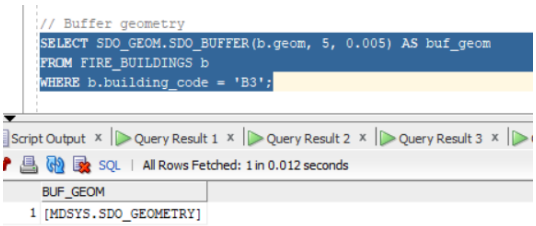
Query 1. Buffer around Observation Tower and list points inside buffer (find POIs within 5 units of the tower)

// // Buffer geometry

SELECT SDO_GEOM.SDO_BUFFER(b.geom, 5, 0.005) AS buf_geom

FROM FIRE_BUILDINGS b

WHERE b.building_code = 'B3';



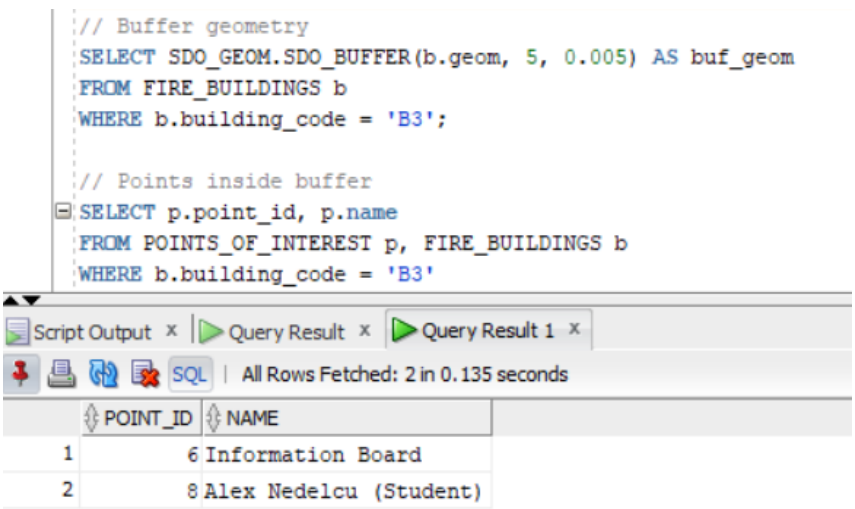
// Points inside buffer

SELECT p.point_id, p.name

FROM POINTS_OF_INTEREST p, FIRE_BUILDINGS b

WHERE b.building_code = 'B3'

AND SDO_WITHIN_DISTANCE(p.geom, b.geom, 'distance=5') = 'TRUE';



Query 2. Centroid and area of each building

SELECT building_code, name,

MDSYS.SDO_GEOM.SDO_AREA(geom, 0.005) AS area,

MDSYS.SDO_GEOM.SDO_CENTROID(geom, 0.005).sdo_point AS centroid

FROM FIRE_BUILDINGS;

```
SELECT building_code, name,
       MDSYS.SDO_GEOM.SDO_AREA(geom, 0.005) AS area,
       MDSYS.SDO_GEOM.SDO_CENTROID(geom, 0.005).sdo_point AS centroid
FROM FIRE_BUILDINGS;
```

Script Output | Query Result 1 | Query Result 2

SQL | All Rows Fetched: 9 in 0.064 seconds

	BUILDING_CODE	NAME	AREA	CENTROID
1	B1	Fire Truck Garage	42	[MDSYS.SDO_POINT_TYPE]
2	B2	Dispatch Center	16	[MDSYS.SDO_POINT_TYPE]
3	B3	Observation Tower	4	[MDSYS.SDO_POINT_TYPE]
4	B4	Administrative Building	1	[MDSYS.SDO_POINT_TYPE]
5	B5	Workshop	16	[MDSYS.SDO_POINT_TYPE]
6	B6	Storage Facility	20	[MDSYS.SDO_POINT_TYPE]
7	B7	Medical Office	12	[MDSYS.SDO_POINT_TYPE]
8	B8	Training Hall	36	[MDSYS.SDO_POINT_TYPE]
9	B9	Alexandru Nedelcu Student Center	25	[MDSYS.SDO_POINT_TYPE]

Query 3. Length of each road

```
SELECT road_id, name,
       MDSYS.SDO_GEOM.SDO_LENGTH(geom, 0.005) AS length
FROM ROADS;
```

Script Output | Query Result 1 | Query Result 2

SQL | All Rows Fetched: 9 in 0.033 seconds

	ROAD_ID	NAME	LENGTH
1	1	Main Alley	30
2	2	Side Road	39
3	3	Boulevard	30
4	4	Access Street	30
5	5	North-South Route	30
6	6	Connector	20
7	7	Bypass	27
8	8	Training Track	12
9	9	Alexandru Nedelcu Avenue	25

Query 4. Nearest point of interest to each building (SDO_NN)

// For building B1, nearest POI

SELECT p.point_id, p.name

FROM POINTS_OF_INTEREST p

WHERE SDO_NN(p.geom, (SELECT geom FROM FIRE_BUILDINGS WHERE building_code='B1'), 'sdo_num_res=1') = 'TRUE';

```
// For building B1, nearest POI
SELECT p.point_id, p.name
FROM POINTS_OF_INTEREST p
WHERE SDO_NN(p.geom, (SELECT geom FROM FIRE_BUILDINGS WHERE building_code='B1'), 'sdo_num_res=1') = 'TRUE';
```

Script Output × | Query Result 1 × | Query Result 2 ×

SQL | All Rows Fetched: 1 in 0.007 seconds

	POINT_ID	NAME
1	5	Bus Stop

-- For all buildings (nearest POI)

SELECT b.building_code, b.name AS building_name, p.point_id, p.name AS poi_name

FROM FIRE_BUILDINGS b

JOIN POINTS_OF_INTEREST p

ON SDO_NN(p.geom, b.geom, 'sdo_num_res=1') = 'TRUE';

```
// For all buildings (nearest POI)
SELECT b.building_code, b.name AS building_name, p.point_id, p.name AS poi_name
FROM FIRE_BUILDINGS b
JOIN POINTS_OF_INTEREST p
  ON SDO_NN(p.geom, b.geom, 'sdo_num_res=1') = 'TRUE';
```

Script Output × | Query Result 1 × | Query Result 2 × | Query Result 3 ×

SQL | All Rows Fetched: 9 in 0.044 seconds

	BUILDING_CODE	BUILDING_NAME	POINT_ID	POI_NAME
1	B1	Fire Truck Garage	5	Bus Stop
2	B2	Dispatch Center	1	Main Entrance
3	B3	Observation Tower	8	Alex Nedelcu (Student)
4	B4	Administrative Building	8	Alex Nedelcu (Student)
5	B5	Workshop	7	Hydrant
6	B6	Storage Facility	4	Secondary Gate
7	B7	Medical Office	6	Information Board
8	B8	Training Hall	2	Fountain
9	B9	Alexandru Nedelcu Student Center	4	Secondary Gate

Query 5. PL/SQL: report with areas, nearest POI, and equipment counts

```
SET SERVEROUTPUT ON SIZE 1000000;
DECLARE
  v_area NUMBER;
  v_nearest_poi VARCHAR2(200);
  v_equip_cnt NUMBER;
BEGIN
  FOR r IN (SELECT building_code, name, geom FROM FIRE_BUILDINGS) LOOP
    v_area := MDSYS.SDO_GEOM.SDO_AREA(r.geom, 0.005);
    -- nearest POI
    SELECT p.name INTO v_nearest_poi
    FROM POINTS_OF_INTEREST p
    WHERE SDO_NN(p.geom, r.geom, 'sdo_num_res=1') = 'TRUE' AND ROWNUM=1;

    -- equipment count
    SELECT COUNT(*) INTO v_equip_cnt FROM EQUIPMENT e WHERE e.building_code = r.building_code;

    DBMS_OUTPUT.PUT_LINE('Building: ' || r.building_code || ' - ' || r.name);
    DBMS_OUTPUT.PUT_LINE('  Area: ' || v_area);
    DBMS_OUTPUT.PUT_LINE('  Nearest POI: ' || NVL(v_nearest_poi,'NONE'));
    DBMS_OUTPUT.PUT_LINE('  Equipment count: ' || v_equip_cnt);
    DBMS_OUTPUT.PUT_LINE('----------------------------------------');
  END LOOP;
END;
/
```

```
    FROM POINTS_OF_INTEREST p
    WHERE SDO_NN(p.geom, r.geom, 'sdo_num_res=1') = 'TRUE' AND ROWNUM=1;

    -- equipment count
    SELECT COUNT(*) INTO v_equip_cnt FROM EQUIPMENT e WHERE e.building_code = r.building_code;

    DBMS_OUTPUT.PUT_LINE('Building: ' || r.building_code || ' - ' || r.name);
    DBMS_OUTPUT.PUT_LINE('  Area: ' || v_area);
    DBMS_OUTPUT.PUT_LINE('  Nearest POI: ' || NVL(v_nearest_poi,'NONE'));
    DBMS_OUTPUT.PUT_LINE('  Equipment count: ' || v_equip_cnt);
    DBMS_OUTPUT.PUT_LINE('----------------------------------------');
  END LOOP;
END;
/
```

Script Output | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4 | Query Result 5

Task completed in 0.173 seconds

```
>>Query Run In:Query Result 4
Building: B1 - Fire Truck Garage
  Area: 42
  Nearest POI: Bus Stop
  Equipment count: 2
----------------------------------------
Building: B2 - Dispatch Center
  Area: 16
  Nearest POI: Main Entrance
  Equipment count: 0
----------------------------------------
Building: B3 - Observation Tower
  Area: 4
  Nearest POI: Alex Nedelcu (Student)
  Equipment count: 1
```

Query 6. Topological relations (touch / overlap / disjoint) between B1 and B2

```
SELECT
 CASE
   WHEN MDSYS.SDO_GEOM.RELATE(b1.geom,'TOUCH',b2.geom,0.005)='TRUE' THEN 'TOUCH'
   WHEN MDSYS.SDO_GEOM.RELATE(b1.geom,'OVERLAPBDYINTERSECT',b2.geom,0.005)='TRUE' THEN 'OVERLAP'
   WHEN MDSYS.SDO_GEOM.RELATE(b1.geom,'DISJOINT',b2.geom,0.005)='TRUE' THEN 'DISJOINT'
   ELSE 'OTHER'
 END AS relation
FROM FIRE_BUILDINGS b1, FIRE_BUILDINGS b2
WHERE b1.building_code='B1' AND b2.building_code='B2';
```

```
SELECT
 CASE
   WHEN MDSYS.SDO_GEOM.RELATE(b1.geom,'TOUCH',b2.geom,0.005)='TRUE' THEN 'TOUCH'
   WHEN MDSYS.SDO_GEOM.RELATE(b1.geom,'OVERLAPBDYINTERSECT',b2.geom,0.005)='TRUE' THEN 'OVERLAP'
   WHEN MDSYS.SDO_GEOM.RELATE(b1.geom,'DISJOINT',b2.geom,0.005)='TRUE' THEN 'DISJOINT'
   ELSE 'OTHER'
 END AS relation
FROM FIRE_BUILDINGS b1, FIRE_BUILDINGS b2
WHERE b1.building_code='B1' AND b2.building_code='B2';
```

Script Output | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4 | Query Result 5

SQL | All Rows Fetched: 1 in 0.009 seconds

	RELATION
1	OTHER

Note: B1 is not a closed polygon, which is why 'other' appears.